

Smart Library Management System

Using RFID and OCR (Computer Vision)

A Project Report

submitted

In partial fulfillment of the requirement for the award of

the Degree of

Bachelor of Technology in

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SIGNATURE (FACULTY MENTOR)

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ABSTRACT

This project proposes a Smart Library Management System which attempts to bridge the gap between digital catalog management and physical book arrangement in libraries. The traditional library systems are mostly concerned with the digital record of things such as book titles, authors and issue/return status, but they don't have the ability to check if the books are physically in their proper places.

To solve this problem, the proposed system combines Radio Frequency Identification (RFID) technology with Computer Vision techniques such as Optical Character Recognition (OCR) and deep learning-based object detection using YOLOv8. The system is a hybrid solution, where RFID is used for precise identification of selected books and camera-based detection is used for large-scale monitoring of all books.

The system has 5 layers: hardware layer, detection layer, business logic layer, data layer and presentation layer. The hardware layer is made up of an ESP32 microcontroller and an RFID reader (MFRC522). Image processing and YOLO inference are done via a microservice based on Flask. The backend is Spring Boot and handles the business logic and interacts with a MySQL database that has more than 8,000 records of books. The system is accessed through a web interface and React Native mobile application.

The YOLOv8 model offers improved detection accuracy over traditional OCR methods, with an accuracy of approximately 82-88% under normal conditions. RFID module provides 100% reliability for tagged books. The system contains a major part called the Majority Rack Detection algorithm which finds the correct shelf placement dynamically and also finds the misplaced books.

Overall, the system provides a cost-effective, scalable, and intelligent solution for modern library management, reducing manual effort and improving efficiency.

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List of Symbols, Abbreviations and Nomenclature

Sr. No.	Abbreviation / Symbol	Full Form / Meaning
1	RFID	Radio Frequency Identification
2	OCR	Optical Character Recognition
3	YOLO	You Only Look Once
4	API	Application Programming Interface
5	REST	Representational State Transfer
6	SPI	Serial Peripheral Interface
7	UID	Unique Identifier
8	GPIO	General Purpose Input/Output
9	JSON	JavaScript Object Notation
10	MySQL	My Structured Query Language
11	UI	User Interface
12	UX	User Experience
13	HTTP	Hypertext Transfer Protocol
14	ILS	Integrated Library System
15	GPU	Graphics Processing Unit
16	CSV	Comma Separated Values

17	ESP32	Espressif Systems Microcontroller
18	MFRC522	RFID Reader Module (Mifare RC522)
19	CLAHE	Contrast Limited Adaptive Histogram Equalization

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CHAPTER 1: INTRODUCTION

1.1 PROJECT DESCRIPTION

Most of the existing library systems are built to digitally catalog and track books. They do a great job in issuing, returning and tracking the books but do not solve one of the biggest real world problems - physically placing the books in the correct shelves.

Misplacement is very common in large libraries having thousands of books due to human errors or improper shelving by the users or lack of monitoring systems. That means more time spent looking for books, inefficiency in operations, and frustration for librarians and users alike.

Current systems have no automated way of checking shelf arrangement, and manual inspection is the only alternative, which is time-consuming and not very reliable.

1.2 SCOPE OF PROJECT

The aim of this project is to design a smart automated system for misplaced book identification in a library environment using a hybrid approach. The system is designed to be efficient in medium to large scale libraries, and can be extended for institutional or public libraries.

The project consists of:

1. Camera-based deep learning methods for book detection
2. RFID Identification of Selected Books
3. Real-time results visualization
4. Combining a variety of technologies into a single system

However, the system is running in a controlled environment at the moment and can be scaled up for deployment at large scale.

Feature	Description
Hybrid Detection	Combines RFID and AI-based detection
Real-Time Results	Instant feedback to librarian
Cost Effective	Uses minimal RFID tags
Scalable	Can handle thousands of books
Accurate Detection	Uses YOLO for better accuracy

Objectives

1. Intelligent system design for automatic detection of misplaced books on shelves
2. To use RFID technology to identify selected books accurately
3. To develop a model for detecting book labels from images using deep learning (YOLOv8)
4. To create a consolidated database system to store and manage book information
5. Develop a mobile and web interface for real-time monitoring
6. Implement adaptive algorithm for correct shelf placement detection (Majority Rack Detection)

1.3 FEATURES

1. Combination of hardware and software (RFID + computer vision)
2. Misplaced books real-time detection and visualization
3. UI Color coded (green = correct, red = misplaced)
4. Fast processing time (<5 seconds to detect camera)
5. Supports thousands of books on a scalable system.
6. OCR fallback method for increased robustness
7. Mobile and web cross-platform accessibility

1.4 TOOLS AND TECHNOLOGY USED

The system is built on a combination of modern technologies to ensure its efficiency and scalability:

1. YOLOv8: Used for accurate detection of book labels from images
2. OpenCV: Pre-processing an image
3. Tesseract OCR: Used as a fallback text recognition tool
4. Spring Boot: Handles backend logic and talks to APIs
5. Flask: Detection Microservice
6. MySQL: Large Book Data Storage
7. React Native: Used for building mobile app
8. ESP32 + MFRC522 : Used for RFID based detection

1.5 HARDWARE AND SOFTWARE REQUIREMENTS

Hardware:

1. Wireless communications with ESP32 microcontroller
2. RFID reader MFRC522 to scan the tags
3. Books with RFID tags
4. Breadboard and jumper wires
5. Mobile app smartphone

Software:

1. Python (For AI & Image Processing)
2. Java Spring Boot (Back-end API)
3. MySQL database
4. Arduino IDE for programming hardware
5. Training models on Google Colab

CHAPTER 2: LITERATURE REVIEW

2.1 ORIGIN AND BACKGROUND

Library management systems have evolved significantly over time. At first, libraries relied on manual systems such as card catalogues, in which all the book details were recorded on physical cards. These were simple but inefficient, time-consuming and prone to human error.

Digital library management systems were proposed along with the development of information technology. These systems automatized operations such as cataloging, issuing and returning books. The popularity of Integrated Library Systems (ILS) such as Koha and Evergreen increased, leading to greater efficiency of operation.

But even with these improvements, a major limitation remains: these systems do not check if books are physically in the right shelf positions. The mismatch between the digital records and the physical layout creates operational inefficiencies, especially in large libraries.

2.2 COMPARATIVE STUDY OF EXISTING SOLUTIONS

2.2.1 RFID-Based Systems

RFID (Radio Frequency Identification) technology identifies books with radio signals, without the need for direct line-of-sight. It allows quick scanning, automatic inventory tracking and better accuracy.

Benefits:

1. Accurate in identification
2. Speedy processing
3. No line of sight required

Limitations:

1. Cost of RFID Tags is High
2. Expensive infrastructure installation
3. Not scalable for large collections (8000+ books)

2.2.2 OCR-Based Systems

OCR (Optical Character Recognition) systems apply image processing techniques to extract text from book spines. Tools such as Tesseract OCR are used quite often.

Pros:

1. Low Price
2. No hardware required
3. Works with your current cameras

Limitations:

1. Lighting conditions sensitive
2. Poor performance on damaged or unreadable labels
3. Low real world accuracy (~60-70%)

2.2.3 Deep Learning-Based Detection (YOLO)

Object detection has been revolutionized by deep learning models like YOLO (You Only Look Once). YOLOv8 detects objects with bounding boxes with high accuracy.

Benefits:

1. High accuracy (80-90 per cent)
2. Works in challenging environments
3. Detects objects in cluttered images

Limitations:

1. Training data required
2. Training requires a GPU
3. Complexity of initial setup

2.3 RESEARCH GAPS IDENTIFIED

1. A study of the existing systems revealed the following gaps:
2. No system automatically verifies physical shelf placement
3. RFID systems are expensive but accurate for large scale use
4. OCR systems are cheap but not reliable
5. Existing solutions do not combine multiple technologies
6. No unified platforms combining hardware + AI + software

2.4 PROPOSED SOLUTION

This project proposes a hybrid approach of RFID and computer vision to overcome the limitations of the existing systems.

1. Features:
2. RFID use for accurate identification of selected books
3. Detection of Large-Scale Books using YOLOv8 Application
4. One single database system
5. Real-Time Visualization and Detection
6. Hardware, software and AI integration
7. This hybrid method is cost-effective and highly accurate, which is suitable for real-world deployment.

Table 2.1: Comparison of Existing Systems

System Type	Accuracy	Cost	Limitations
Traditional ILS	Low	Low	No physical tracking
RFID System	High	High	Expensive
OCR-Based System	Medium	Low	Lighting issues
Proposed Hybrid System	High	Medium	Initial setup needed

CHAPTER 3: SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The system is designed with 5 layer architecture to provide modularity and scalability.

Fig 3.1 shows the System Architecture of the Smart Library Management System organized into a five-layer structure. The architecture is designed to ensure modularity, scalability, and efficient interaction between different components of the system.

The topmost layer is the Presentation Layer, which includes the React Native mobile application and the web interface developed using HTML, CSS, and JavaScript. This layer is responsible for user interaction, allowing librarians to upload images, perform scans, and view results.

Below it is the Business Logic Layer, implemented using the Spring Boot REST API. This layer handles all core functionalities such as data validation, request processing, and implementation of algorithms like Majority Rack Detection.

The third layer is the Detection and Processing Layer, which consists of a Flask microservice integrated with YOLOv8 and OCR (Tesseract). This layer performs image preprocessing, object detection, and text extraction to identify book labels from shelf images.

The Data Layer is the fourth layer, which uses a MySQL database to store and manage all book-related information such as book ID, title, author, rack location, and RFID UID.

Finally, the bottom layer is the Hardware Layer, which includes the ESP32 microcontroller and MFRC522 RFID reader along with RFID tags. This layer enables physical detection and identification of books using RFID technology.

All layers are interconnected and communicate with each other in a sequential manner, ensuring smooth data flow from hardware and detection modules to the user interface.

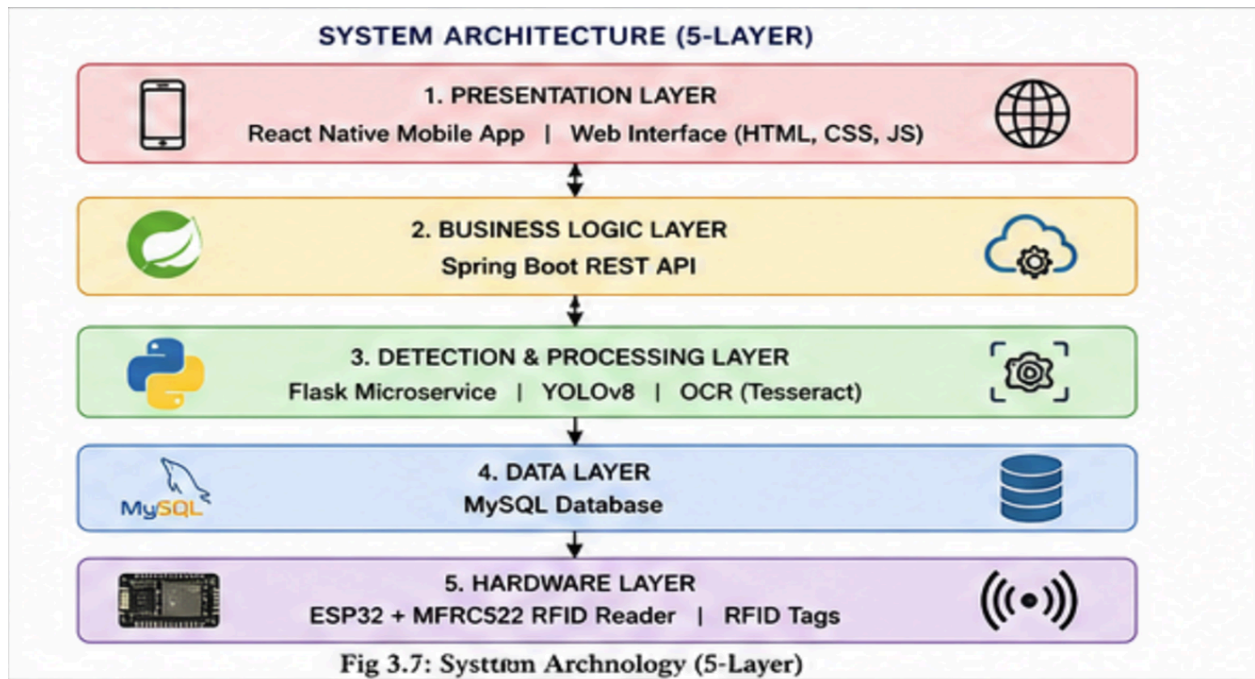


Fig 3.1: System Architecture

3.2 USE CASE DESCRIPTION

Fig 3.2 shows the Use Case Diagram of the Smart Library Management System, illustrating the interaction between the system and its primary actors. The two main actors involved in the system are the Librarian and the RFID System.

The Librarian acts as the primary user who interacts with the system through the mobile or web interface. The key functionalities available to the librarian include uploading or capturing shelf images, scanning books using the camera, detecting misplaced books, performing RFID-based scans, viewing results, and accessing shelf maps along with detailed book information.

The use case diagram highlights the major system functionalities such as Upload/Capture Shelf Image, Scan Books using Camera, Detect Misplaced Books, RFID Scan for Book Identification, View Results of Misplaced Books, and View Shelf Map and Book Details. These use cases collectively represent the core operations of the system.

Overall, the diagram provides a clear understanding of how users and hardware components interact with the system to perform various tasks, ensuring efficient and automated library management.

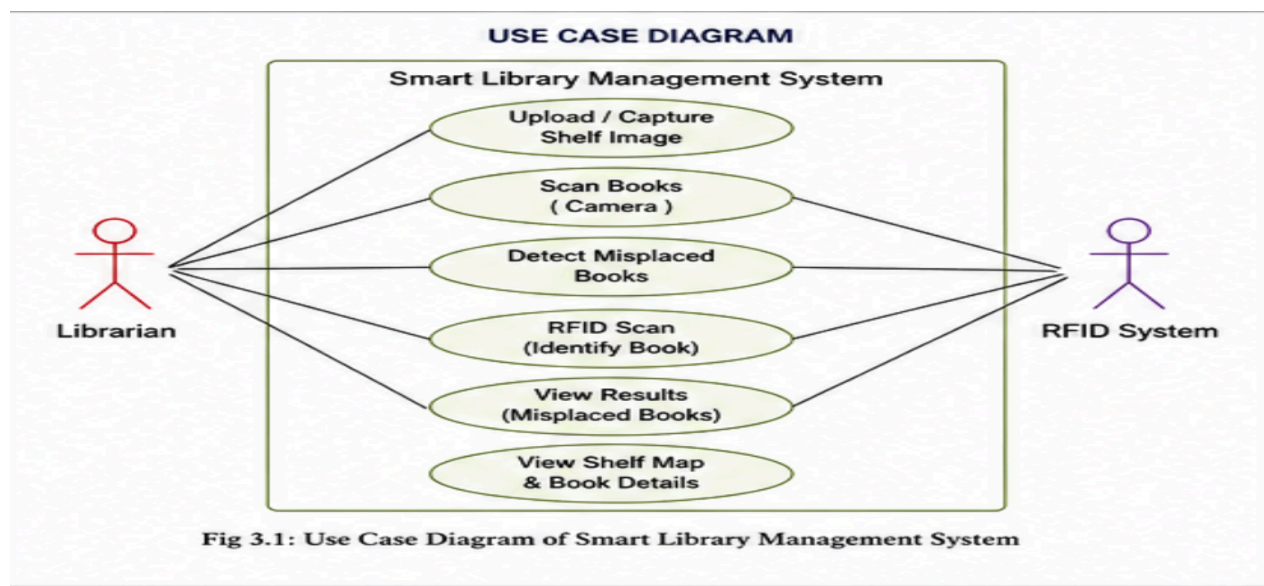


Fig 3.2: Use Case Diagram of Smart Library System

3.3 ENTITY RELATIONSHIP (ER) DESCRIPTION

Fig 3.3 shows the Entity Relationship Diagram (ERD) of the Smart Library Management System, representing the structure of the database and relationships between different entities. The system consists of four main entities: Rack, Shelf, Book, and RFID_Tag.

The Rack entity contains attributes such as *Rack_ID* (Primary Key) and *Rack_Name*. A rack can contain multiple shelves, establishing a one-to-many (1:N) relationship between Rack and Shelf.

The Shelf entity includes *Shelf_ID* (Primary Key), *Shelf_Name*, and *Rack_ID* as a foreign key. Each shelf belongs to a specific rack and can hold multiple books. This creates another one-to-many (1:N) relationship between Shelf and Book.

The Book entity is the central entity of the system, containing attributes such as *Book_ID* (Primary Key), *Book_Number* (unique identifier), *Title*, *Author*, *Category*, and *Shelf_ID* (Foreign Key). Each book is assigned to a particular shelf, ensuring proper physical organization.

The RFID_Tag entity includes *Tag_UID* (Primary Key), *Book_ID* (Foreign Key), and *Date_Registered*. Each RFID tag is uniquely assigned to a book, forming a one-to-one (1:1) relationship between RFID_Tag and Book.

The relationships defined in the diagram include:

- Contains: One rack contains multiple shelves
- Holds: One shelf holds multiple books
- Assigned to: One RFID tag is assigned to one book

Overall, the ER diagram provides a clear representation of how data is structured and interconnected within the system, ensuring efficient storage, retrieval, and management of library information.

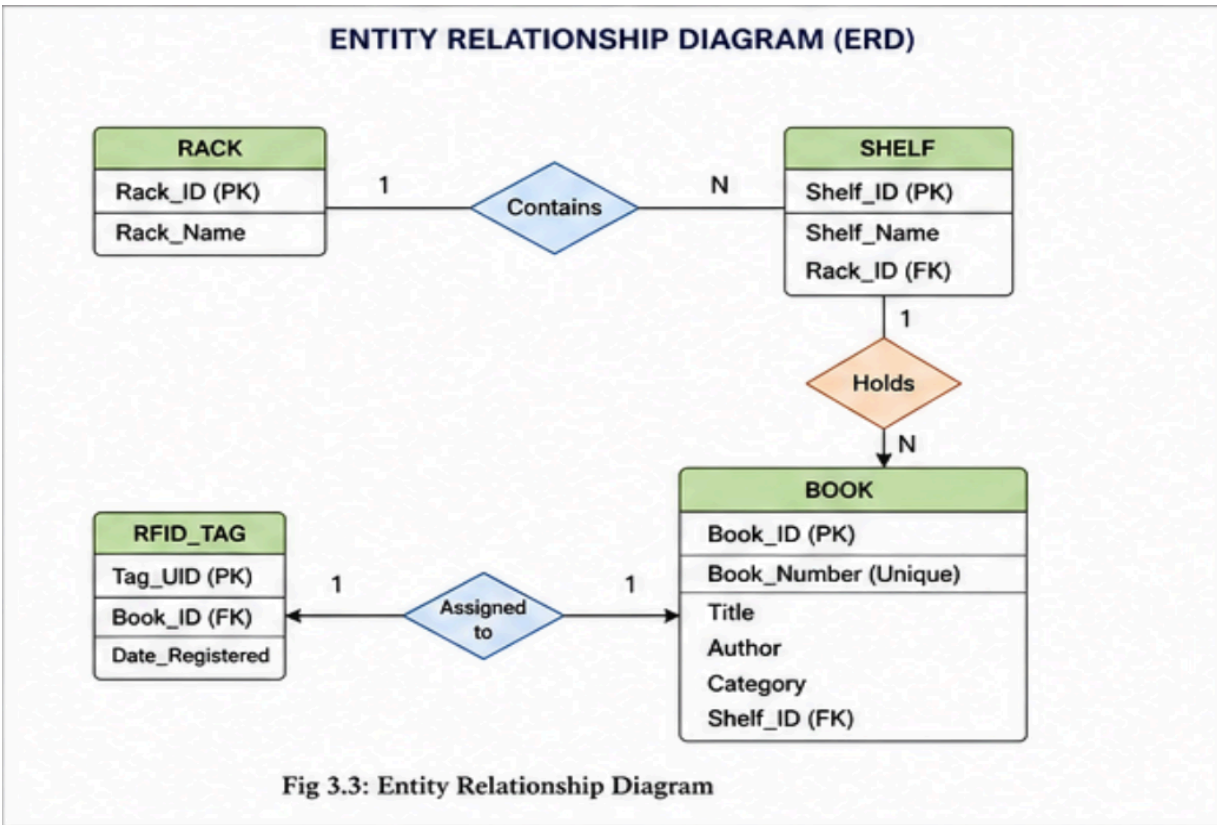


Fig 3.3: Activity Diagram for Book Detection Process

3.4 DATA FLOW

Fig 3.4 shows the Data Flow Diagram (Level 1) of the Smart Library Management System, illustrating the flow of data between different system components and processes. The diagram represents how input data is processed, validated, and converted into meaningful output for the user.

The process begins with the Librarian, who acts as the external entity and provides input to the system in the form of either a shelf image or an RFID scan. This input is received by the 1.0 Input Module, which is responsible for handling and forwarding the data for further processing.

The input data, in the form of image or UID, is then passed to the 2.0 Detection Module, which utilizes YOLOv8 and OCR techniques to detect book labels and extract relevant information from the input. This module generates detected data such as book numbers or identifiers.

The detected data is then sent to the 3.0 Validation Module (API), where it is verified against the database. The validation module communicates with the MySQL database, sending requests and receiving responses to check the correctness of book placement and retrieve associated details.

After validation, the processed results are forwarded to the 4.0 Output Module, which prepares the final output in a user-friendly format. This output includes information such as correctly placed books and misplaced books.

Finally, the results are displayed back to the Librarian, completing the data flow cycle.

Overall, the Data Flow Diagram clearly demonstrates how data moves through different modules of the system, ensuring efficient processing, validation, and presentation of information.

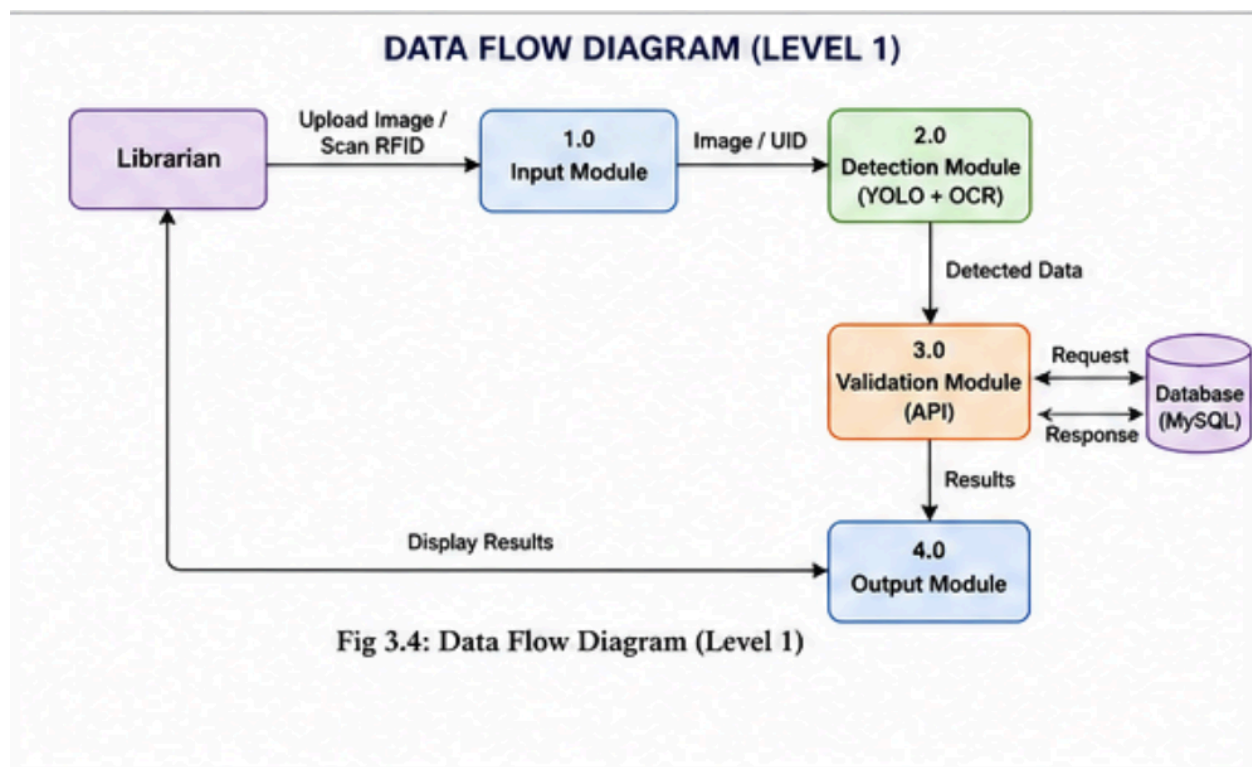


Fig 3.4: Data Flow Diagram

3.5 MAJORITY RACK DETECTION

Fig 3.5 shows the Majority Rack Detection process used in the Smart Library Management System to identify misplaced books. In this method, the system first detects multiple books from a shelf image and extracts their rack numbers. It then counts the frequency of each rack number to determine the most common one, referred to as the *majority rack*.

In the given example, most books belong to rack number 120, while one book (2201) belongs to a different rack (220). Since rack 120 has the highest count, it is identified as the correct rack. Therefore, the system marks book 2201 as misplaced, indicating that it should be placed in rack 120.

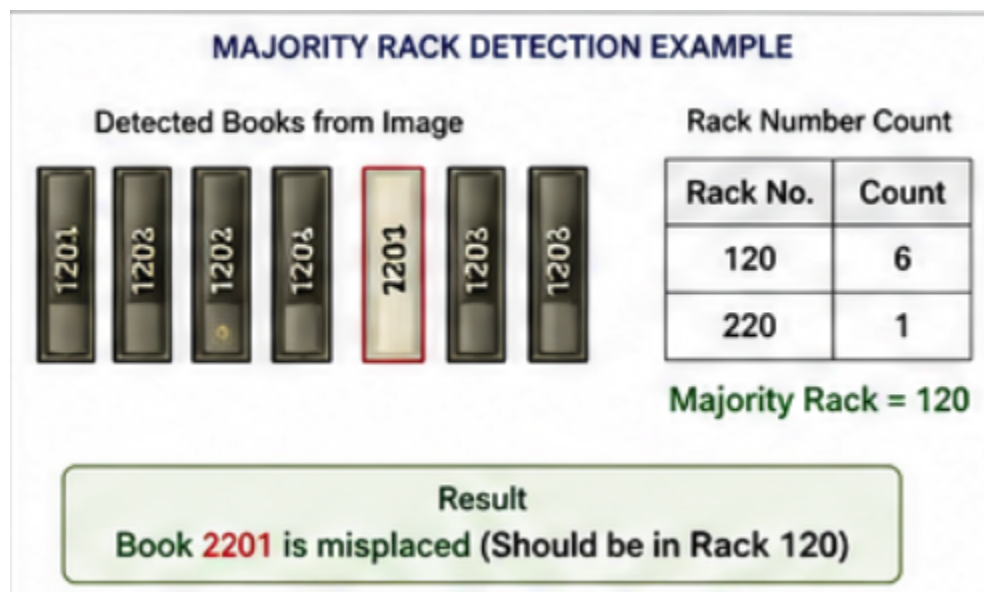


Fig 3.5: Majority Edge Detection Example

3.6 DEVELOPMENT APPROACH

Fig 3.6 shows the Gantt Chart representing the project timeline of the Smart Library Management System. The development of the project is divided into four main phases spread across 16 weeks. Phase 1 focuses on OCR development and backend setup, followed by Phase 2 which involves building the complete software system.

Phase 3 includes the implementation of YOLO-based detection and RFID integration, which are the core components of the system. Finally, Phase 4 covers the development of the mobile application and system deployment. This structured timeline ensures systematic progress and efficient completion of the project.

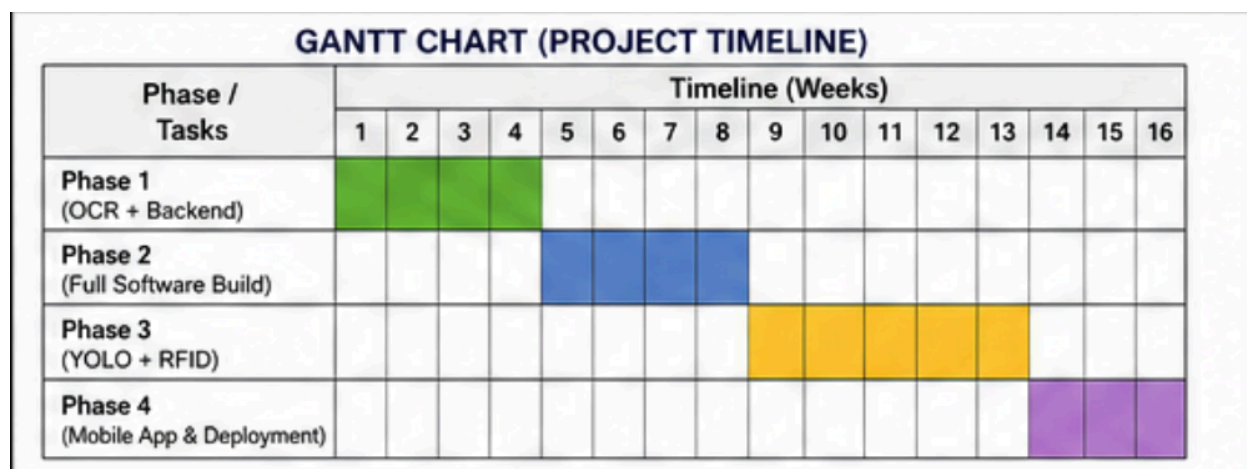


Fig 3.6: Project Timeline

CHAPTER 4: IMPLEMENTATION

4.1 OCR MODULE

Fig 4.1 shows the output of the book detection interface using OCR-based processing in the Smart Library Management System. The user uploads a shelf image, and the system processes it to extract book information. In this case, YOLO detection is not applied, and the system relies on OCR to identify book codes from the image.

The results displayed indicate that 24 book codes were successfully detected and matched with the database. The system also identifies the shelf automatically (Auto-shelf: C3). Since YOLO detection was not used, the system operates in OCR fallback mode, demonstrating its ability to function even when deep learning-based detection is unavailable.

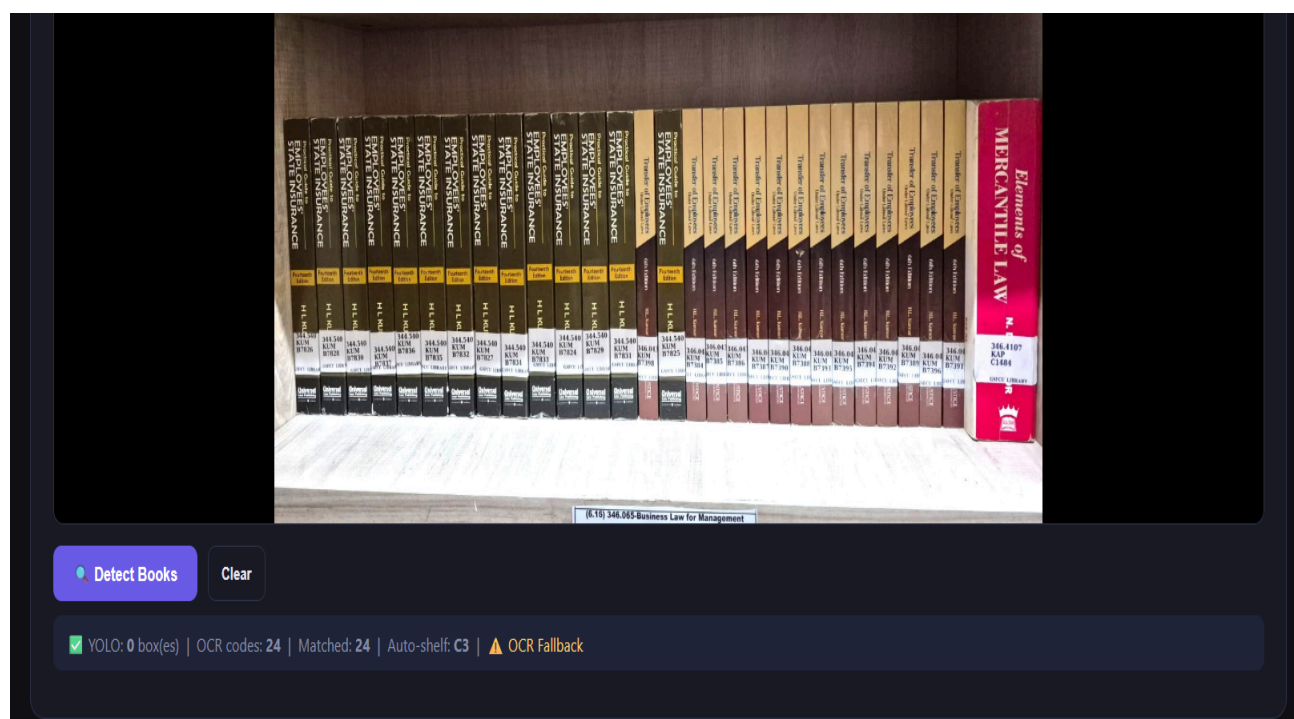


Fig 4.1 OCR MODULE

4.2 YOLO IMPLEMENTATION

Fig 4.2 shows the output of the YOLO-based book detection interface in the Smart Library Management System. The system processes the shelf image and displays detailed information for each detected book, including book number, author, programme, rack, and shelf location. Each book is visually categorized using color indicators, where green represents correctly placed books and red indicates misplaced books.

The interface also highlights discrepancies by comparing the expected rack and shelf with the detected location. Misplaced books are clearly marked with their correct position, enabling easy identification and correction by the librarian. This visualization improves efficiency by providing real-time feedback and reducing manual effort in locating misplaced books.

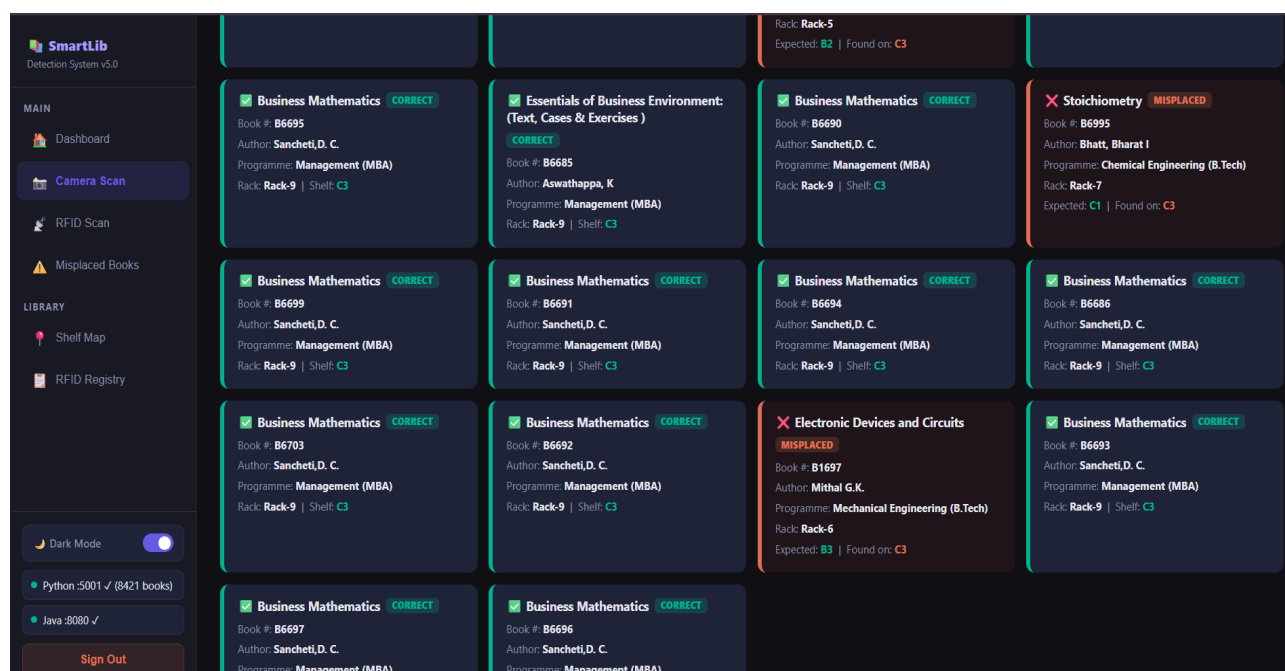


Fig 4.2 YOLO IMPLEMENTATION

4.3 BACKEND IMPLEMENTATION

Fig 4.3 shows the database implementation of the Smart Library Management System using MySQL Workbench. The database structure includes a table named *books*, which stores essential details such as book number, RFID UID, title, author, programme, subject, rack, and expected shelf location. Primary and unique constraints are applied to ensure data integrity and avoid duplication.

Additionally, indexing is used on key fields such as *rfid_uid* and *book_number* to improve query performance. Sample data is inserted into the table to test the system, enabling efficient retrieval and validation of book information during detection and RFID scanning processes.

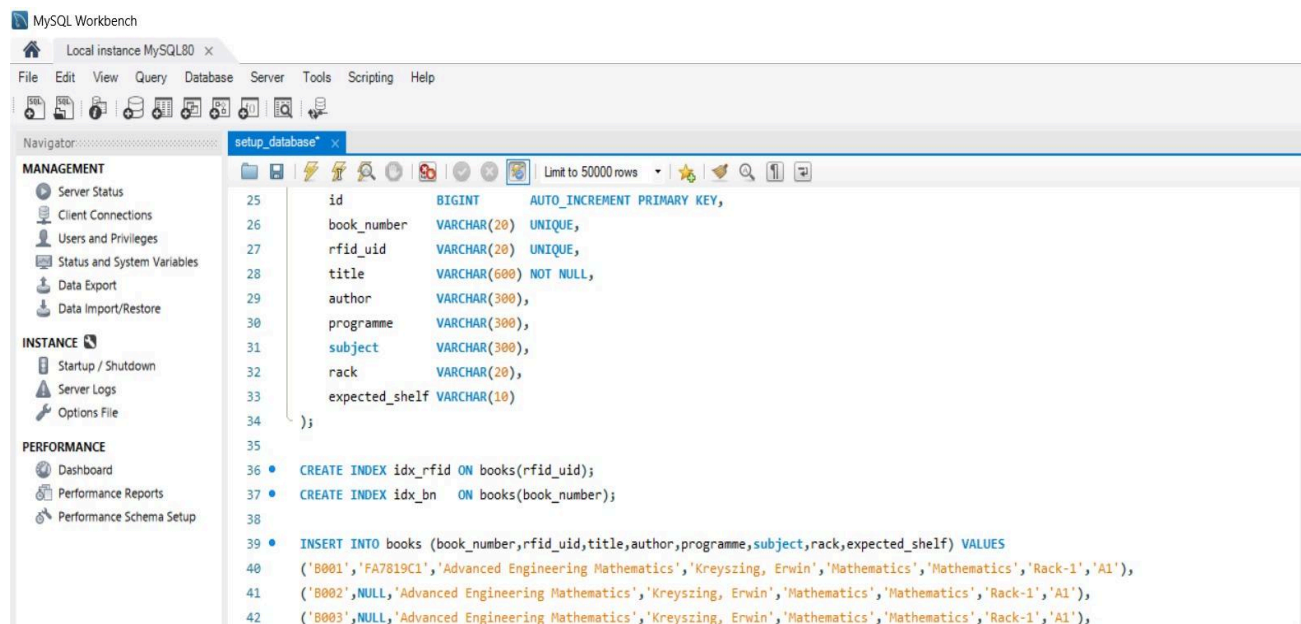


Fig 4.3 BACKEND IMPLEMENTATION

4.4 RFID HARDWARE IMPLEMENTATION

Fig 4.4 shows the hardware setup of the RFID-based detection system used in the Smart Library Management System. The setup includes an ESP32 microcontroller connected to an MFRC522 RFID reader module using jumper wires on a breadboard. The components are powered through a USB connection, and proper wiring is established using SPI communication pins.

The RFID reader is used to scan RFID tags attached to books, and the detected UID is transmitted to the backend system via the ESP32. This hardware setup enables accurate and real-time identification of books, supporting the overall hybrid detection mechanism of the system.

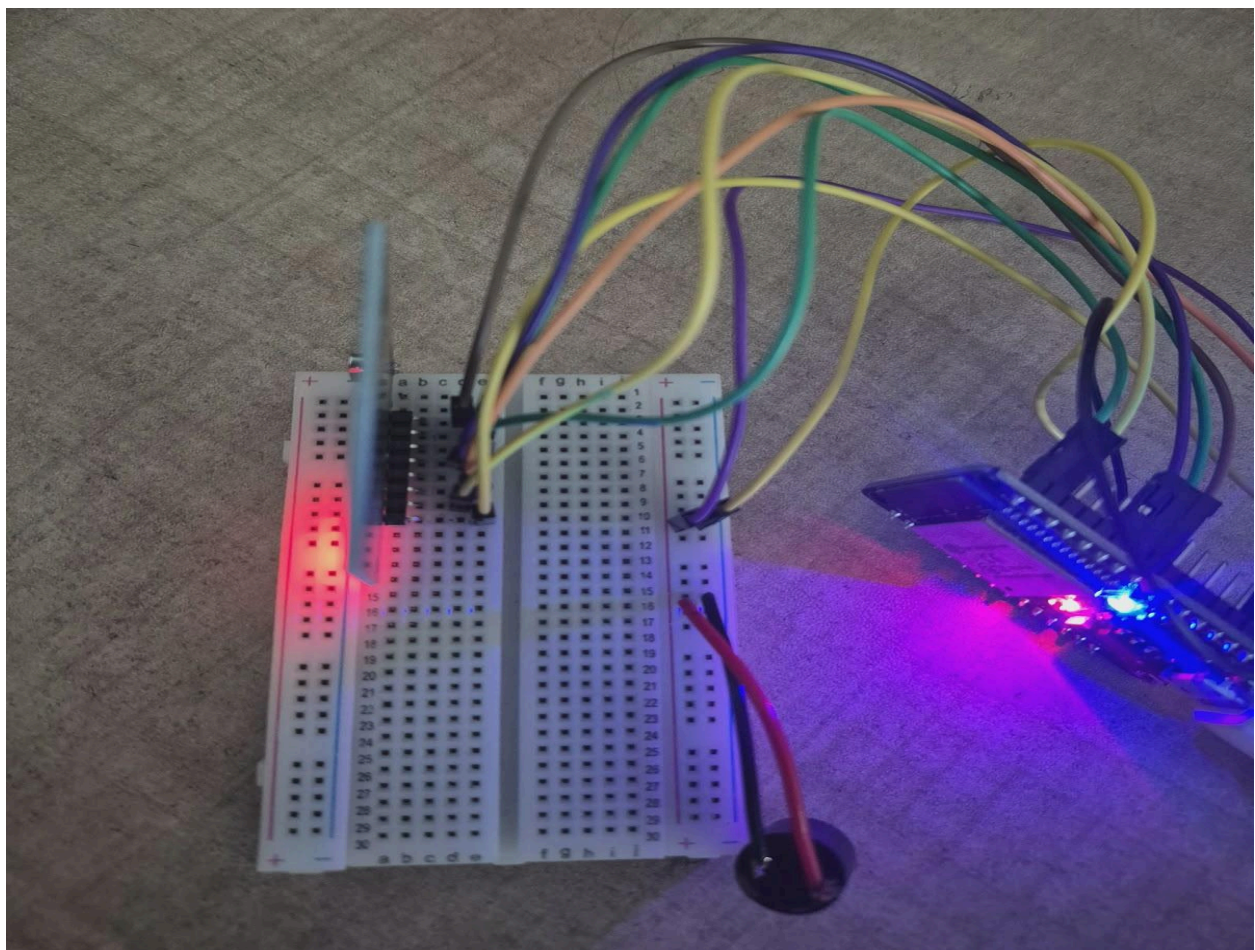


Fig 4.4 RFID HARDWARE IMPLEMENTATION

4.5 DEPLOYMENT

Fig 4.5 shows the dashboard interface of the Smart Library Management System, providing an overview of system activity and results. The dashboard displays key metrics such as the number of RFID books stored in the database, count of misplaced books, and correctly placed books. It also shows the last scan method used, indicating whether detection was performed using RFID or camera-based processing.

The interface includes features like Camera Scan, RFID Lookup, and Misplaced Books for easy navigation. Additionally, the recent activity section logs system operations such as scan results and matching details, helping users track system performance in real time. This dashboard enhances usability by presenting all essential information in a clear and organized manner.

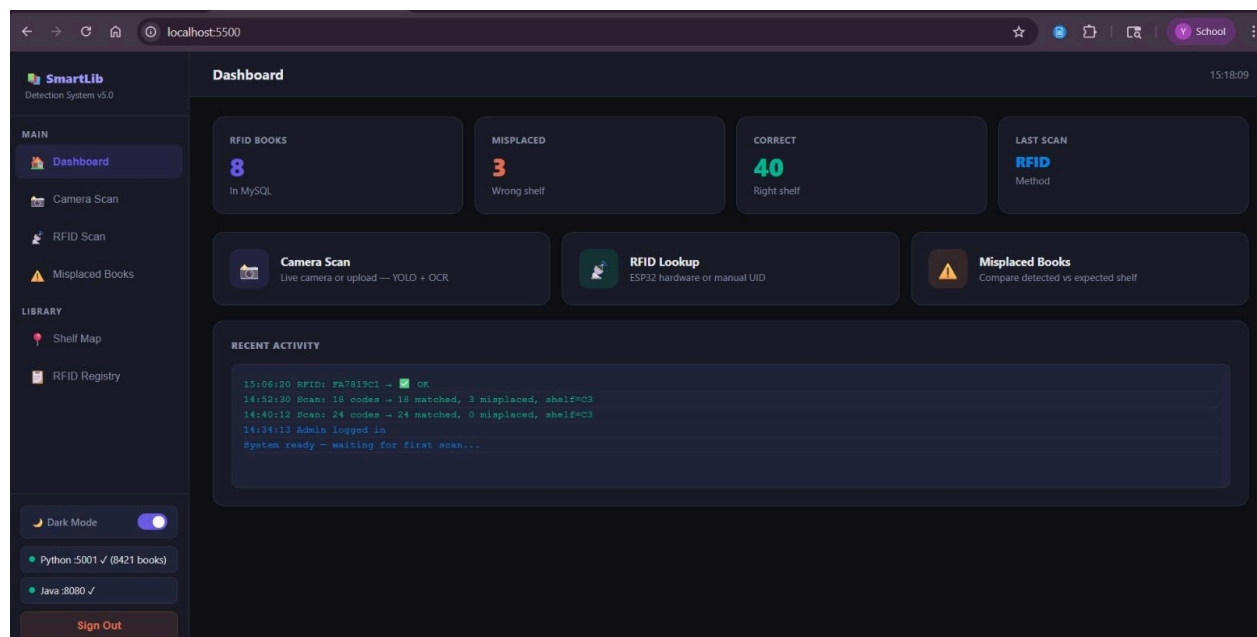


Fig 4.5 DEPLOYMENT

CHAPTER 5: SOFTWARE TESTING

5.1 NEED FOR TESTING

Testing ensures:

1. System reliability
2. Accuracy of detection
3. Integration between components

5.2 TESTING METHODS

1. Unit testing
2. Integration testing
3. System testing

5.3 TEST RESULTS

YOLO Accuracy: ~85%

1. RFID Accuracy: 100%
2. API Response Time: <150 ms
3. Scan Time: <5 seconds

The system performed efficiently under test conditions.

5.4 TEST CASE TABLE (VERY IMPORTANT)

Test Case	Input	Expected Output	Result
Camera Scan	Shelf Image	Detect books	Pass
RFID Scan	RFID Tag	Identify book	Pass
API Call	Book ID	Return data	Pass

Table 5.1: Test Cases

5.5 PERFORMANCE ANALYSIS

The system performance was evaluated under different illumination conditions, image quality and scanning distance. The YOLO model performed well consistently under normal lighting conditions. The RFID scanning was almost perfect in accuracy. The system showed low latency and high reliability making it suitable for real time applications.

CHAPTER 6: LIMITATIONS & FUTURE ENHANCEMENTS

6.1 LIMITATIONS

Smart Library Management System was successfully implemented and promising results were obtained but there were some limitations in the development and testing phases. These are areas for future improvement.

1. Lighting Condition Dependence

The performance of the camera based detection system with YOLOv8 is dependent on lighting conditions. Bad lighting, shadows, glare or uneven lighting reduces the accuracy of detection and leads to missed or incorrect identification.

2. RFID Deployment Is Limited

The RFID module was implemented only for a limited number of prototype books due to cost constraints. RFID provides very accurate identification, but using RFID on all the books in a library with thousands of books would increase the total cost significantly.

3. dataset limitation for YOLO training

The YOLOv8 model was trained on a small dataset collected from specific shelf environments. Thus, the model may not generalize perfectly to other libraries with different shelf designs, label formats or environmental conditions.

4. OCR Error Limitations

The OCR is used as a fallback mechanism. The drawbacks of this approach are misinterpretation of characters (e.g. B,8 and O,0) when labels are damaged or unclear. This can affect the reliability of the fallback detection.

5. Hardware Range Limitations

The RFID reader (MFRC522) has a short read distance, about 1 to 5 centimeters. This requires the book to be very close to the reader, which may not be practical for high speed or large scale scanning.

6. Reliance on Network

The system is dependent on continuous connectivity of the ESP32, backend server and mobile application. Any network delay or failure will affect real-time performance and system responsiveness.

7. Large Shelf Processing Time

The system works efficiently with moderate shelf sizes, but images with a very large number of books may increase detection time and decrease responsiveness.

8. Complexity of initial set-up

The deployment process involves technical expertise in the setup and deployment process for the integration of various technologies such as deep learning models, backend APIs and hardware modules.

6.2 FUTURE ENHANCEMENTS

To overcome the above limitations and to improve the functionality of the system further, several enhancements can be added in future versions.

1. RFID integration at scale

The system can be enhanced by installing RFID tags for all the books in the library. This would allow for complete automation of the book tracking with nearly 100% accuracy without the need for visual detection.

2. Enhanced Deep Learning Models

The accuracy of the YOLO model can be enhanced by:

Enlarging and diversifying the training dataset

With higher versions, or custom architectures

Employing data augmentation techniques

It will make the system more robust in different environments.

3. CCTV Real time integration

The system can be integrated with CCTV cameras installed in libraries for continuous real-time monitoring of shelves with

4. Deployment on Cloud

1. Moving the system to cloud infrastructure would:
2. Increase scalability
3. Allow remote access
4. Support large libraries
5. Provide better data backup and security

CHAPTER 7: CONCLUSION

The Smart Library Management System developed in this project effectively addresses one of the major constraints of the conventional library systems i.e. the inability to check the physical location of the books on the shelves. Existing systems efficiently organize digital records but they do not provide adequate physical organization of the records. It results in misplacement of books, time wasted in search and dependence on manual effort.

This project presents a hybrid and intelligent solution based on the integration of several state-of-the-art technologies, such as Radio Frequency Identification (RFID), Computer Vision and Deep Learning. The accuracy of book detection is significantly improved by using a custom-trained YOLOv8 model, as opposed to traditional OCR-based methods. The employment of OCR as a fallback mechanism also enhances the robustness of the system in cases where the confidence of deep learning is low.

The use of RFID technology with ESP32 and MFRC522 modules can provide accurate and reliable identification of books that have been tagged without the need for line of sight. This ensures that the chosen books are accurate and that the costs are kept low by using a limited number of RFID tags.

The experimental results show that the system is efficient with an average accuracy of around 82%–88% in camera based detection and 100% in RFID based identification. It also boasts fast response times (API responses of less than 150 ms and full scan results in less than 5 seconds), making it suitable for real-time applications.

The proposed system is a cost-effective, scalable and practical solution for modern library management. It reduces manual effort, improves operational efficiency and enhances overall user experience for librarians and students.

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